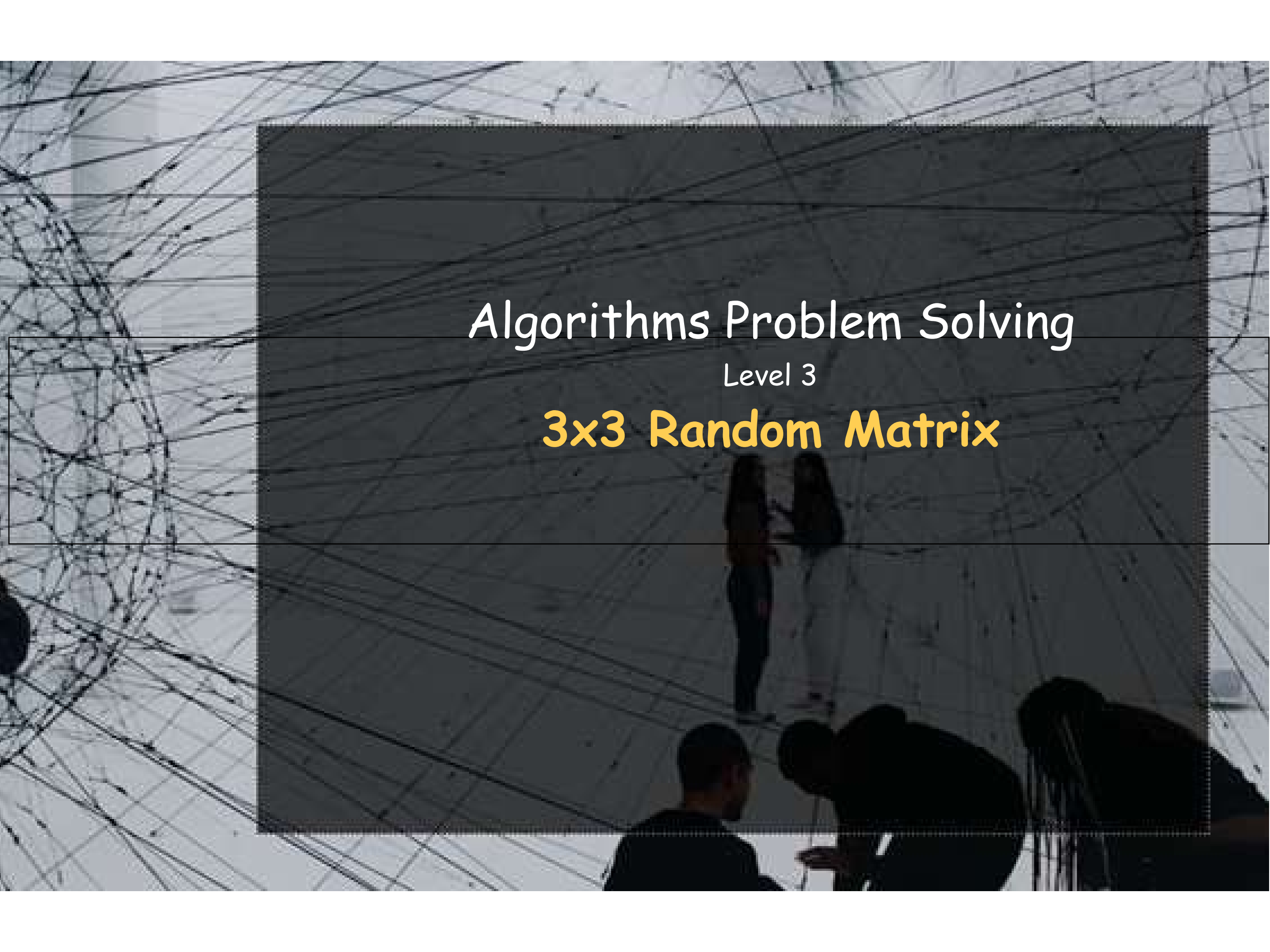


Algorithms Problem Solving

Level 3

3x3 Random Matrix



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Definition

3x3 Random Matrix

-  The goal is to create a 3x3 matrix and fill it with random numbers between 1 and 100.
-  The matrix is then displayed on the screen in a formatted table.

Explanation

2D `vector`

 We use a two-dimensional `vector` to store the matrix values.

`rand()`

 `rand()` generates pseudo-random numbers.

`srand(time(0))`

 `srand(time(0))` changes the random sequence based on the current time.

 This helps produce different sequences when the program is run at different times.

`rand() % 100 + 1`

 `% 100` produces a value from 0 to 99.

 Adding 1 changes the range to 1 through 100.

`setw(3)`

 `setw(3)` from `*iomanip*` formats the output by reserving a width of 3 characters for each number.

Procedures

1 Define the matrix:

```
vector<vector<int>> matrix(3, vector<int>(3));
```

 Creates a 2D vector containing 3 rows and 3 columns.

2 Generate random numbers:

 Inside `FillMatrixWithRandomNumber()`, initialize the random number generator:

```
srand(time(0));
```

 Fill each element with a random number:

```
rand() % 100 + 1
```

3 Print the matrix:

 Inside `PrintMatrix()`, use nested loops to print the rows and columns.

 Use `setw(3)` from `*<iomanip>*` to align the numbers.

4 Call the functions from `main()`:

```
FillMatrixWithRandomNumber(matrix, 3, 3);
```

```
PrintMatrix(matrix, 3, 3);
```

 First fill the matrix with random numbers, then print it.

💡 Example

The following is a 3x3 random matrix:

16 13 24

45 51 19

30 3 100

➡ Result

Each time the program runs, the matrix can contain different random values between 1 and 100.

Note

- ◆ ``srand(time(0))`` should normally be called once at the beginning of the program, not every time a random number is generated.
- ◆ ``rand()`` generates pseudo-random numbers, not truly random numbers.

⊘ Warning

- ◆ Calling `srand(time(0))` repeatedly in a short period can produce the same sequence because `time(0)` may have the same value.
- ◆ In modern C++, `<random>` with engines such as `std::mt19937` is generally preferred over `rand()` and `srand()` for better control and randomness quality.



Comparison

▨ ``rand()` + `srand()``

- ✓ Simple and commonly used in beginner exercises.
- ✓ Easy to understand.
- ✓ Uses ``srand(time(0))`` to change the sequence between runs.
- ✓ Older C-style approach.

▤ ``<random>``

- ✓ Modern C++ random-number facilities.
- ✓ Provides better control over random engines and distributions.
- ✓ Generally preferred in modern C++ programs.

✦ Summary

- ✓ Create a 3x3 matrix using `vector<vector<int>>`.
- ✓ Use `rand()` to generate pseudo-random numbers.
- ✓ Use `rand() % 100 + 1` to generate values from 1 to 100.
- ✓ Use `srand(time(0))` to initialize the random sequence.
- ✓ Use nested loops to fill and print the matrix.
- ✓ Use `setw(3)` from `<iomanip>` to format the output.

Thank you 😊